DATA-DRIVEN SOLUTIONS AND DISCOVERIES IN MECHANICS USING PHYSICS-INFORMED NEURAL NETWORKS

Machine Learning Mini-Project • Implementation and Results Summary

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1. Problem Statement

This project implements Physics-Informed Neural Networks (PINNs) for mechanics problems governed by differential equations. A conventional neural network learns from labelled examples; a PINN also incorporates governing physics, initial conditions, and boundary conditions into its training loss. The project studies three cases: one-dimensional consolidation, two-dimensional steady heat conduction, and inverse identification of damping and stiffness in a damped single-degree-of-freedom (SDOF) system.

2. Dataset and Problem Details

No external dataset file is required. Training points are generated from the physical domains, boundary/initial conditions, and—only for the inverse problem—synthetic displacement observations generated from a known oscillator. The implementation uses PyTorch and automatic differentiation to compute derivatives required by the differential equations.

3. Methodology and Implementation

PINN training principle. The neural network predicts a physical field (pressure, temperature, or displacement). Automatic differentiation computes its derivatives, and the residual of the governing equation becomes part of the loss. The optimizer adjusts network weights to reduce physics and condition errors. In the inverse experiment, unknown physical parameters are trainable variables as well.

Experiment 1 — 1D consolidation

The pressure field is modelled with the diffusion equation:

$$\partial p/\partial t - \partial^2 p/\partial x^2 = 0, \quad x \in [0, 1], \quad t \in [0, 0.5]$$

The implementation applies p(0,t)=0 through the output transformation p(x,t)=x·N(x,t), and penalizes the remaining initial/boundary conditions in the loss. The predicted field is compared with an analytical series solution. Paper-mode training uses 50,000 Adam epochs followed by L-BFGS refinement.

Experiment 2 — 2D steady heat conduction

The temperature field satisfies the Poisson equation:

$$\partial^2 T/\partial x^2 + \partial^2 T/\partial y^2 + 1 = 0, \quad (x,y) \in [-1,1]^2; \quad T = 0 \text{ on the boundary}$$

The model is trained using interior collocation points and boundary points. Since this notebook does not compute an independent finite-element or analytical temperature reference, evaluation reports the PDE residual on a test grid and the maximum predicted boundary magnitude.

Experiment 3 — Inverse damped dynamics

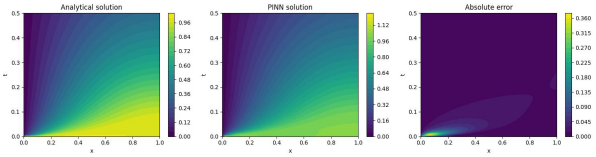
The governing equation is $m \cdot u'' + c \cdot u' + k \cdot u = 0$, with m=1 and synthetic observations generated using c=0.4 and k=4.0. The neural network predicts displacement u(t), while c and k are learned jointly with the network. The observation interval [0,10] is an implementation choice; it is not specified in the paper excerpt. Paper-mode training uses 100,000 Adam epochs.

4. Results and Discussion

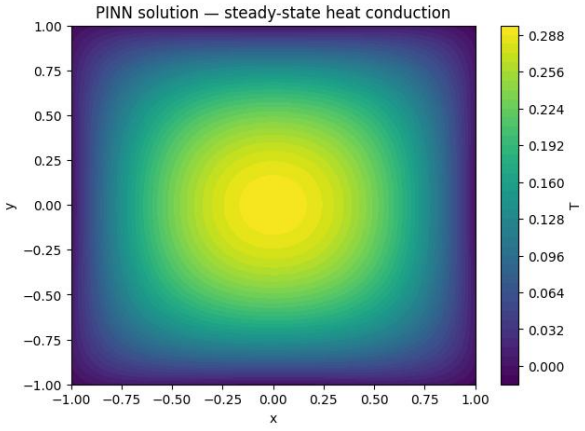
The following values are taken from the saved outputs of the paper-mode notebook.

Experiment	Training / evaluation	Saved result	Interpretation
1. Consolidation	50,000 Adam epochs + L-BFGS	MSE = 5.8467×10^{-4} Max |error| = 0.36229	The PINN and analytical fields broadly agree. The largest error is localized near the initial/boundary corner; the maximum error is more noticeable than the overall MSE alone suggests.
2. Heat conduction	50,000 Adam epochs + L-BFGS	Test-grid PDE residual MSE = 2.3214×10^{-6} Max boundary |T| = 0.011878	The temperature field is smooth and peaks near the centre. The small residual indicates good PDE consistency at the test points; it is not a direct temperature error against an independent reference.
3. Inverse dynamics	100,000 Adam epochs	c: 0.400803 (true 0.4) k: 3.993249 (true 4.0)	The identified parameters are close to the synthetic ground-truth values. Relative errors are approximately 0.20% for c and 0.17% for k.

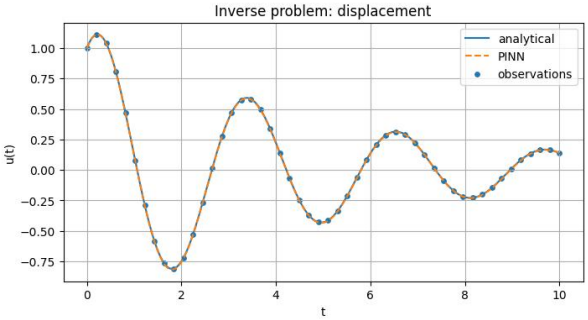
Selected implementation figures



Experiment 1: analytical field, PINN field, and absolute error.



Experiment 2: learned steady temperature field.



Experiment 3: displacement fit against synthetic observations.

5. Conclusion and Limitations

The experiments demonstrate that a PINN can approximate a field governed by a differential equation and infer unknown physical parameters when supplied with observations. The inverse oscillator recovered c and k accurately in this run. The consolidation result has a low mean-squared error but a larger localized maximum error, so both metrics should be reported. The heat experiment shows a small test-grid PDE residual and boundary deviation; however, without an independent temperature reference, these checks do not establish the absolute solution error. Results can depend on collocation-point sampling, initialization, network capacity, and optimization. L-BFGS closure-evaluation counts alone should not be interpreted as a formal convergence guarantee.

6. Reference

“Data Driven Solutions and Discoveries in Mechanics Using Physics Informed Neural Network.” Source paper supplied for the mini-project. Implementation and numerical results are from the project notebook notebooke81568c9f1.ipynb.